

# Improving large language model assisted categorization and classification of highway construction accidents

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## ABSTRACT

The construction industry, particularly in highway work zones, grapples with persistently high rates of occupational injuries and fatalities, requiring a transformative approach to bolster safety measures. This study advances current research in safety by harnessing modern technologies, including natural language processing and machine learning. The primary objective presents a systematic, data-driven approach for future understanding and mitigation of accidents, using text-rich narratives extracted from publicly available Occupational Safety and Health Administration (OSHA) databases. The approach incorporates classical clustering algorithms, with large language models (LLMs) playing a pivotal role in classification, comprehension, and dissemination. A MapReduce summarization process is introduced to characterize high-level commonalities between similar incidents, achieving 83% fact consistency by effectively filtering hallucinations and noisy data. The results demonstrate the efficacy of LLMs in automating safety data management; while zero-shot classification showed moderate performance, few-shot prompting achieved high precision across eight classification tasks, with certain categories such as “degree of injury” and “human factor” exceeding 94% accuracy. Manual validation of the “event type” field further revealed that in 37.7% of miscoded cases, the model identified more contextually relevant classifications than the original database. Anchored in OpenAI’s Generative Pre-trained Transformer (GPT-3.5), the study positions LLMs as a robust foundation for automated safety analyses and data quality control in construction or other industries with similar concerns.

## 1. Introduction

Recent data indicates a high incidence of accidents within the construction industry, with multiple cases reported worldwide (Ayhan and Tokdemir, 2020; Zhou et al., 2023). This has resulted in a substantial number of workers being fatally injured at construction sites, with approximately 108,000 workers losing their lives annually (Shanti et al., 2021). Despite its recognition as a high-risk sector, the fatality rate in construction has steadily increased in the U.S. over the past decade (Zhou et al., 2023). One in ten construction workers in the U.S. sustains injuries each year (Shanti et al., 2021), many of which occur in the highway industry. These metrics mirror the industry’s lack of systematic identification of precursors to fatalities, accidents, and near-misses (Fang et al., 2020). Though existing studies have touched on safety aspects within highway construction work zones, there remains an opportunity to develop more comprehensive approaches to identify and

quantify leading types of accidents in this area.

Common work zone hazards in highway construction, such as falling from heights and being struck by objects, are widespread and reveal unique safety challenges in the industry (Liu et al., 2023). Risks related to man/machine interface and the fall of materials, particularly during work at height, significantly contribute to incidents (Dhalmahapatra et al., 2019). The presence of harmful objects, like misused tools, nearby sharp objects, and damaged equipment, also poses risks (Zhang et al., 2019). The multitude of hazards underscores the urgent need for improved safety measures and hazard recognition in the construction industry (Fang et al., 2020; Do et al., 2022; Uddin et al., 2023).

To enhance safety within work zones, a comprehensive data-driven approach that draws insights from accident analyses is essential. This involves examining accident records to understand past incidents and mitigate future risks (Do et al., 2022). Additionally, Uddin et al. advocate for revising educational and training programs to better equip

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future construction professionals for workplace safety (Uddin et al., 2023). Deng et al. stress the importance of analyzing direct accident causes, particularly focusing on equipment components and failure causes (Deng et al., 2020). These studies promote the development of new models to address safety hazards, potentially reducing costs, fatalities, and injuries on construction sites (Ayhan and Tokdemir, 2020). While these methods have prevailed in other domains, limited applications have been extended to highway construction.

With a wealth of data on existing accidents, researchers and practitioners are empowered to make safety interventions (Ayhan and Tokdemir, 2020), learning from historical occurrences. Employing techniques like natural language processing (NLP) in accident analysis provides a deeper understanding of incidents and facilitates the development of preventative measures (Zhang et al., 2019; Chokor et al., 2016; Tian et al., 2023). Modern text processing methods excel in classifying causes, types of injuries, and affected body parts, significantly advancing safety knowledge and improving safety plans (Do et al., 2022). These techniques can effectively extract accident causes, establish connections between narratives and causes, and support safety inspections, with application extending to many disciplines such as chemical security (Deng et al., 2020; Feng et al., 2021), mining safety (Pothina and Ganguli, 2023), and railway safety (Heidarysafa et al., 2018).

NLP in construction safety is explored in various applications using Term Frequency – Inverse Document Frequency (TF-IDF) and Latent Dirichlet Allocation (LDA), which primarily rely on numerical representations of texts (Weng et al., 2022). Provided the challenge of performing computation on raw text, these techniques serve as a foundation for advanced analysis. TF-IDF is a numerical statistic that reflects the importance of word frequency within a text corpus. In contrast, LDA is a statistical model that assigns probabilities to words across different topics (Baker et al., 2020; Lossio-Ventura et al., 2021). When coupled with other machine learning (ML) algorithms, such as clustering or classification, these tools can effectively analyze textual narratives of accidents in the construction industry (Zhou et al., 2023; Dhalmahapatra et al., 2019; Zhang et al., 2019; Chokor et al., 2016; Baker et al., 2020; Shehadeh and Alshboul, 2025).

These methods identify dominant words in predefined categories, making them suitable for classification and topic modeling tasks. For example, Chokor et al. combine TF-IDF and K-means clustering to classify accident reports in Arizona (Chokor et al., 2016). The results reveal key terms belonging to four main categories: falls, struck by objects, electrocutions, and trench collapse. However, with no extension to other regions, and the limited information available after analysis, it is difficult to transfer key terms into actionable safety proceedings.

Similarly, Zhang et al. applied text mining and NLP techniques through various ML algorithms including support vector machine, linear regression, K-nearest neighbor, decision tree, and Naïve Bayes (Zhang et al., 2019). The results show a variation in performance between these advanced models, all of which require extensive preprocessing. The authors emphasize that manual postprocessing of the analysis is still required, even though their suggested methodology does not require labeled data (Zhang et al., 2019). Additionally, they highlight the difficulty in dealing with natural language, specifically how easy it is to misinterpret the intended context.

The introduction of language models like Bidirectional Encoder Representations and Transformers (BERT) has permitted further enhancement of the automatic classification of textual accident data within safety reports (Fang et al., 2020). Such models offer a significant advancement over traditional term frequency-based methods by mapping the context of words in a sentence while considering the surrounding words (Weng et al., 2022). Many studies incorporate these mappings, or embeddings, to facilitate a more comprehensive capture of information in complex construction safety texts (Fang et al., 2020; Liu et al., 2023; Tian et al., 2023; Wang and Zhao, 2022). Aside from BERT, other common embedding models such as GloVe and word2vec have

been used for automatic text classification of construction accident reports (Luo et al., 2023). Moreover, when integrated with other text processing and ML methods, they demonstrate high performance in classifying data within safety reports (Fang et al., 2020; Danish et al., 2025), highlighting the potential of deep learning methods in enhancing the accuracy and efficiency of safety analyses.

Though language models have shown themselves to be superior to more traditional NLP approaches (Pothina and Ganguli, 2023), their accuracy is limited to the size of the training data. Novel large language models (LLMs), exemplified by OpenAI's Generative Pre-trained Transformer (GPT), represent a new frontier in computer science and technology. These models are extensively trained on vast datasets to acquire syntactic, semantic, and world knowledge, enabling them to perform over a wide range of tasks. Functioning as general-purpose computers, they execute programs specified by natural language prompts, demonstrating abilities across various NLP tasks (Wei et al., 2022; Zhou et al., 2023). Similar to BERT, their implementation with the Transformer architecture, which introduces context awareness through self-attention mechanisms, underpins their effectiveness (Muennighoff et al., 2023). However, GPT models surpass BERT in their ability to generate coherent and contextually relevant text due to their larger training datasets and more sophisticated architecture (Wei et al., 2022; Muennighoff et al., 2023). These state-of-the-art models position themselves as useful in analyzing text with minimal data processing and computer science expertise, prompting them to be convenient for field personnel—especially where automation of labor-intensive tasks may be important (Zhang et al., 2024).

Smetana et al. showcases a recent application of various NLP and ML techniques to analyze textual safety data in the highway construction domain (Smetana et al., 2024). In this study, a novel approach was developed with a strong emphasis on utilizing LLMs to facilitate the identification of significant accidents in the industry. Sourced from the Occupational Safety and Health Administration (OSHA) Severe Injury Reports (SIR) database, 1032 accidents were clustered into six distinct groups using unsupervised K-means. OpenAI's GPT-3.5 model was guided to perform summarization and classification over the entire database, resulting in unique insights that were otherwise difficult to attain with the database alone. While demonstrating an example of safety analysis, the study has notable limitations. The selected SIR database lacks fatality records, which restricts the scope of the analysis. In unsupervised clustering, issues such as the curse of dimensionality and the randomness of centroid initialization are not addressed, impacting robustness. Additionally, some generated summaries are overly general, limiting their practical applicability. Lastly, the classification approach does not evaluate the language model's transfer learning capabilities.

This paper proposes an extension of the approach outlined in the previous study (Smetana et al., 2024), aiming to overcome some of its inherent limitations, ultimately providing the industry with an improved analysis of safety data. This study also employs GPT-3.5 to assist in identifying major accident types within highway construction. By including fatality data, the analysis reveals accident types that differ slightly from those presented in SIR. The new database highlights the dominance of “struck-by” accidents and “falls,” suggesting a more realistic severity of these incidents, as they are identified as fatal rather than just severe injuries. Conversely, injuries that do not typically result in fatalities, such as amputations, are less prevalent in the new data.

Our primary goal is to incorporate several significant improvements to the approach and then test its robustness with new data, ultimately providing a data-driven analysis with insights that extend beyond general NLP and ML capabilities. While traditional NLP methods like TF-IDF or LDA focus on identifying dominant keywords or broad topics, this study introduces a MapReduce summarization process. This approach enables an LLM to synthesize large clusters of unstructured text into a concise list of facts without losing critical technical detail or context, overcoming overly general summaries. Each list of facts is then

evaluated for self-consistency, removing hallucinated or noisy incident details, resulting in coherent, focused summaries that reveal broader commonalities between incidents. By showcasing the potential of LLMs to conduct similar studies in the future, we offer an improved framework to better understand accidents specific to the highway industry or other industries with similar concerns. This methodology is designed to be highly applicable and cost-effective for field personnel by using pre-trained foundation models, reflecting a general shift toward natural language programming rather than specialized code associated with conventional NLP pipelines. With minimal manual evaluation, practitioners can utilize this methodology to pinpoint industry-specific concerns or transfer it to other databases that rely on textual entries.

In summary, the main contributions of this paper can be outlined as follows.

1. The introduction of a MapReduce prompting technique to break down large amounts of unstructured data into smaller chunks to maintain high-level technical detail and context that standard generative methods often lose.
2. Mitigation of LLM to hallucination by a specific self-consistency evaluation step. Each synthesized list of facts is cross-verified to remove noisy incident details, ensuring that the final summary is both coherent and grounded in the actual data.
3. Demonstration of data quality through the transfer learning capabilities of LLMs. By using few-shot prompting for classification, the model identifies miscoded incidents, providing a mechanism for automated reassessment of existing database coding.

## 2. Data and methodology

### 2.1. OSHA IMIS database

The database utilized in this study is the OSHA Integrated Management Information System (IMIS), which contains approximately 100,000 workplace accident cases from various industries since 1972. Of the various datasets found within the IMIS database, the enforcement data was further selected for analysis due to its wealth of textual information. This dataset includes entries from prior inspections, violations, or on an as-event-occurred basis which are mandatorily reported per OSHA's requirements. Compared to the Severe Injury Reports (SIR) database used in a previous study (Smetana et al., 2024), which only required the reporting of hospitalizations, this data also contains incidents that resulted in fatalities. Otherwise, both of these databases contain rich but diverse information about the context of each accident, including some insight into potential causes and the nature of the incidents.

To identify incidents specific to highway construction, the dataset was sorted and filtered. The IMIS database adheres to the North American Industry Classification System (NAICS), where code 237310 is designated for Highway, Street, and Bridge Construction, covering a wide range of activities from pavement rehabilitation to airport runway and bridge construction. The highway construction sector ranks tenth in reported incidents within the IMIS database, comprising 1.87% of the total. This places it behind major contributors such as commercial building construction, roofing contractors, and others. Nevertheless, the significant number of distinct incidents in highway construction, totaling 1198 cases, underscores the inherent dangers and the need for further investigation.

The IMIS database contains extensive details for each reported incident, including information about the involved employers, the age of the injured worker, and the specific time and date of the incident location. Additionally, it encompasses categorical variables that provide information on the nature and consequences of each accident, as well as overarching textual abstracts that offer more event-specific details. Table 1 provides an overview of the categorical information analyzed in this study, including the field names as entered in the database, brief

**Table 1**  
OSHA IMIS database description and top entries.

| Column                      | Description                         | Example Codes                                                                                                                                                                | Top 3 Database Entries (NAICS 237310)                                                                                     |
|-----------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| "event_type"<br>14 codes    | Classification of injury            | struck by, caught in or between, bite/sting/scratch, fall (same level), fall (from elevation), struck against, rubbed/abraded, inhalation, ingestion, absorption, ...        | Struck By (51.75 %)<br>Caught in or Between (17.95 %)<br>Fall (From Elevation) (12.85 %)                                  |
| "nature_of_inj"<br>22 codes | Nature of injury                    | amputation, asphyxia, bruise/contusion/abrasion, burn (chemical), burn/scald (heat), concussion, cut/laceration, dermatitis, dislocation, electric shock, ...                | Other (38.06 %)<br>Fracture (27.55 %)<br>Bruise/Contusion/Abrasion (7.85 %)                                               |
| "part_of_body"<br>31 codes  | Part of body involved in injury     | abdomen, arms multiple, back, body system, chest, ears, elbows, eyes, face, fingers, feet/toes/ankles, hands, head, hips, knees, legs, lower arms, lower legs, multiple, ... | Multiple (20.45 %)<br>Body System (19.53 %)<br>Head (15.94 %)                                                             |
| "env_factor"<br>18 codes    | Potential environmental factor      | pinch point action, catch point/puncture action, shear point action, squeeze point action, flying object action, overhead moving and/or falling object action, ...           | Other (34.89 %)<br>Working surface/Facility Layout Condition (15.19 %)<br>Materials Handling Equipment/Method (13.36 %)   |
| "hum_factor"<br>20 codes    | Potential human factor in incident  | misjudgment of hazardous situation, malfunction of procedure for securing operation or warning of hazardous situation, distracting actions by others, ...                    | Misjudgment of Hazardous Situation (32.55 %)<br>Other (26.96 %)<br>Operational Position Not Appropriate for Task (7.76 %) |
| "occ_code"<br>518 codes     | Occupational code of injured worker | demonstrators, promoters, models., sales support, n.e.c., administrative support, including clerical., supervisors, general office., supervisors, ...                        | Nonclassifiable (39.73 %)<br>Construction laborers (20.12 %)<br>Paving, surfacing, tamping equipment operator (5.26 %)    |
| "src_of_injury"<br>48 codes | Potential external source of injury | motor vehicle (highway), machine, materials handling equipment, drugs/alcohol, hoisting apparatus, working surface, hand tool (powered), motorcycle, bodily motion, ...      | Motor Vehicle – Highway (28.05 %)<br>Motor Vehicle – Industrial (12.77 %)<br>Machine (10.02 %)                            |
| "degree_of_inj"<br>3 codes  | Degree of injury (severity)         | fatal, hospitalized, and non-hospitalized                                                                                                                                    | Fatal (62.6 %)<br>Hospitalized (32.3 %)<br>Non-Hospitalized (4.26 %)                                                      |

descriptions, examples of typical entries, and top entries per field specific to the highway construction industry. These columns were selected as the most useful for aiding in the identification of the underlying causes of incidents.

The “event\_type” column categorizes incidents into typical classifications, indicating the nature of the accident, such as struck-by, caught in or between, falls, etc. Meanwhile, “nature\_of\_inj” describes the actual harm sustained by an employee during an incident. The “part\_of\_body” specifies the injured body part. “evn\_factor” and “hum\_factor” offer insights into potential environmental or human factors contributing to the accident, respectively. The “occ\_code” identifies the type of operational position the employee held during the incident, and “src\_of\_injury” indicates a potential external source contributing to the injury/fatality. Finally, the “degree\_of\_inj” indicates whether the accident resulted in fatality, hospitalization, or no hospitalization. This information is valuable for safety analysis as it provides a comprehensive breakdown of the key factors surrounding each incident, enabling a focused examination of the contributing elements, injury details, and contextual factors.

However, the information derived from the raw data, summarized in Table 1, underscores the limitations inherent in traditional descriptive statistics. While overarching statistics, such as the 62.6% fatality rate, offer a broad perspective, the diverse combinations of entries across multiple columns only provide narrow insights into specific and niche accident categories and their causes. For instance, while there are 14 possible event types from the IMIS coding dictionary, there are also 22 possibilities for the nature title, 31 for the part of the body, and so on. Examining this interplay of various factors reveals only fine-grained information, such as the finding that 29.5% (183) of incidents classified as “struck-by” in the highway construction industry also result in fractures, with an additional 16.4% (30) of those cases affecting the legs. Despite recognizing the prominence of these derived statistics, extracting actionable information for practical application in the field becomes challenging.

In contrast, the abstracts accompanying each incident, sourced from the Accident Investigation OSHA-170 form, offer a wealth of information that may proceed with the other coded entries in the database. These abstracts provide potential sources and outcomes of accidents, presenting a more cohesive picture. Before the advent of advanced NLP techniques, this information may have been underutilized as each abstract varies greatly in details from a single sentence to multiple paragraphs.

## 2.2. Methodology overview

This paper introduces a revised methodology for extracting practical insights from existing accident reports, with a specific focus on highway construction activities. The outlined approach (Fig. 1) can be adapted to textual data in diverse industries, rendering it suitable for both public

and private datasets with similar characteristics.

To initiate the approach, the OSHA incident reporting database was utilized due to its wealth of textually relevant entries, encompassing pre-coded categorical data and contextually rich accident narratives. Next, textually relevant information gets isolated from the dataset, concentrating on individual narrative abstracts that can be used to identify broader similarities between accidents. This deliberate focus leverages the advanced capabilities of LLMs and their ability to understand the context within a large amount of text and generate human-like responses.

Following data preparation, each incident abstract undergoes conversion into numerical vector representations, known as sentence embeddings. An embedding model computes high-dimensional vectors, allowing for integration into traditional ML techniques while preserving semantic meaning. With contextually relevant vectors for each incident abstract, the K-means algorithm is then applied to automatically cluster incidents based on similarities. The algorithm uses a distance-based function to iteratively group incidents, with closely positioned incidents representing major categories of accidents. Following clustering, an advanced dimensionality reduction, t-SNE (t-Distributed Stochastic Neighbor Embedding) (van der Maaten and Hinton, 2008), is applied to visualize high-dimensional embeddings in more perceivable two-dimensional plots, aiding in interpreting an appropriate number of categories.

Once an optimal number of clusters is determined, the language model is employed to execute two main tasks through prompting: summarization and classification. The model is guided to summarize commonalities in each group or cluster, revealing refined representations of major accident categories. Finally, the model is prompted to assess the original coding of the other categorical variables in the database based on context within the previously isolated narrative. In an alternative task, the model attempts to assign categories to these fields without knowledge of the original coding.

## 2.3. NLP and incident embedding representation

The conversion of words into numeric values is imperative for advanced computation, classification, and clustering (Bailey and Chopra, 2018). Traditional methods like term frequency-inverse document frequency (TF-IDF) fall short in capturing word similarity, contextual order, and semantic relations (Baker et al., 2020; Agarwal et al., 2019; Qiu and Yang, 2021). While methods like TF-IDF and other term frequency-based approaches are important for many text classifications approaches, they tend to have inherent limitations and selectivity in weighting and normalization schemes (Chen et al., 2021). Thus, modern approaches in text processing address these limitations through the use of word embeddings like Word2Vec, GloVe, and FastText, enhancing accuracy in classification and text mining applications (Worth, 2023; Singh et al., 2022; Yi et al., 2020; Treistman et al., 2022).

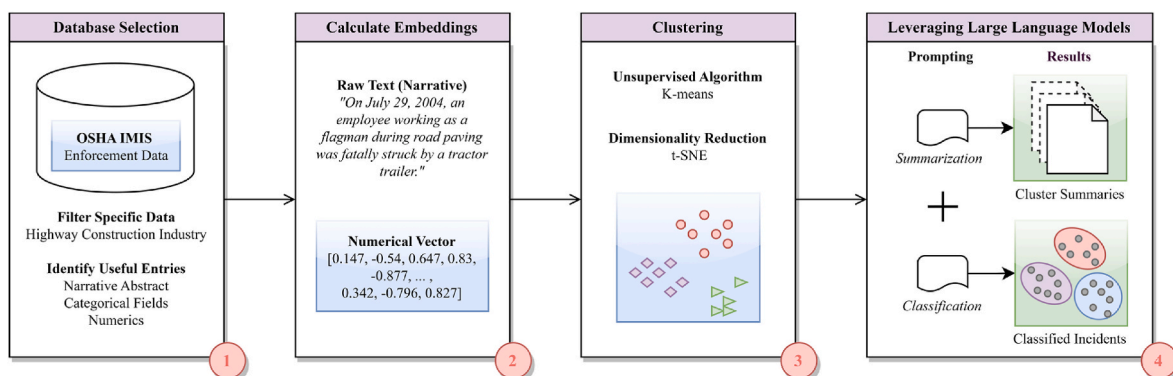


Fig. 1. Overview of proposed approach for analysis of road construction incidents.

Essentially, word embeddings represent words as numerical vectors, capturing semantic relationships between other input words (Treisman et al., 2022). In the realm of construction safety, consider the term “fall protection.” Word embeddings would represent “fall protection” as a vector, encapsulating the semantic relationships associated with this concept. This vector would encode the contextual meaning, capturing the interconnectedness of terms related to fall protection, such as harness, lanyard, OSHA regulations, and safety measures. By doing so, word embeddings provide a nuanced understanding of the semantic landscape within construction safety.

The selection of an adequate embedding model can significantly affect the quality of capturing semantic correlations in input texts. To maintain the focus on clustering and classifying accidents in highway construction, OpenAI's text-embedding-ada-002 pre-trained embedding model is explored. These embeddings have demonstrated outstanding performance in the Massive Text Embedding Benchmark (MTEB), which assesses the capabilities of emerging embedding models across various benchmark datasets (Muennighoff et al., 2023). Since ada-002 ranked high under this intense benchmarking, it was considered suitable for the subsequent tasks applied in this study. Further studies may explore the performance of various embedding models, as the comparative evaluation between different embedding models, including the effect of embedding dimension, were not included in this investigation.

The employed model outperforms other available word embedding models by effectively handling out-of-vocabulary (OOV) words with subwords and capturing word context through bidirectional transformers, which serve as the foundation of many advanced language models (Feng et al., 2021; Asyaky and Mandala, 2021). Ada-002 avoids the computational cost of training and eliminates the need for extensive text training data, enforcing high applicability across diverse datasets (Weng et al., 2022; Banerjee et al., 2023), such as the IMIS investigated in this study. Leveraging the selected model's inherent capabilities, no further pre-training or fine-tuning is performed.

To illustrate how pre-training typically works in other contexts, the pre-training phase involves employing the Transformer Encoder  $E$ . This component maps multiple input sentences, denoted as  $x$  and  $y$ , into high-dimensional vectors  $v_x$  and  $v_y$ , respectively (Neelakantan et al., 2022). This transformation facilitates its ability to capture semantic relationships and contextual information. During this phase, sentences are deliberately structured for effective training. Special start-of-sentence (SOS) and end-of-sentence (EOS) are introduced and concatenated using the symbol  $\oplus$  (Neelakantan et al., 2022). These tokens play a crucial role in delineating the boundaries of sentences, aiding the model in understanding the contextual nuances within the input text.

$$v_x = E([SOS]_x \oplus x \oplus [EOS]_x) \quad (1)$$

$$v_y = E([SOS]_y \oplus y \oplus [EOS]_y) \quad (2)$$

The resulting high-dimensional vectors,  $v_x$  and  $v_y$ , serve as valuable representations that encapsulate the semantic meaning of the corresponding sentences, enabling the grasp of intricate linguistic patterns. These pre-trained models extend beyond word-level similarity, accommodating mapping across one or multiple sentences (Fig. 2), aligning with the structure of IMIS textual abstracts. For instance, in the domain of construction safety, consider a scenario where a worker injures a finger while untangling crane lines. Here, the context exceeds the mere definitions of the words involved. It involves a complex interplay of physical actions, potential hazards, and contextual nuances specific to the construction site environment. This highlights how embeddings capture not just the literal meanings of words, but also the nuanced relationships and contextual subtleties they convey.

To further illustrate this embedding process, as shown in Fig. 2, tokenization through the cl100k\_base tokenizer breaks down each input abstract into smaller chunks (conceptually represented as arbitrary colors). These chunks are then input into text-embedding-ada-002, generating vectors of 1536 dimensions that preserve the contextual information of the original narratives. The left-hand side of this figure demonstrates the step-wise procedure of converting raw text to embeddings, connected by dashed lines to an example of each step.

#### 2.4. Clustering algorithm with high-dimensionality

Unsupervised text clustering methods leverage cluster analysis to synthesize information and extract underlying structures (Balbi, 2010). The commonly used K-means clustering approach categorizes accident data based on spatial information, capturing spatial correlation (Ma et al., 2021). K-means clustering has been employed to identify accident severity locations, revealing prone areas on curved and sloped surfaces (Yassin, 2020). Chokor et al. proposed a K-means based clustering approach to rearrange accidents based on types (Chokor et al., 2016).

In the current study, the K-means algorithm clusters the calculated embeddings from the IMIS database, potentially revealing hidden relationships between highway construction incidents. Calculated embeddings maintain the context of their original text input, enabling the algorithm to assess Euclidean distances ( $d$ ) in the  $n$ -dimensional space from cluster centroids to individual data points ( $p, q$ ), as shown in Eq. (3), through an iterative process.

$$d(p, q) = \sqrt{(p_1 - q_1)^2 + \dots + (p_i - q_i)^2 + \dots + (p_n - q_n)^2} \quad (3)$$

Visualizing datasets of substantial magnitude and high dimensionality in a perceivable two or three-dimensional space demands advanced

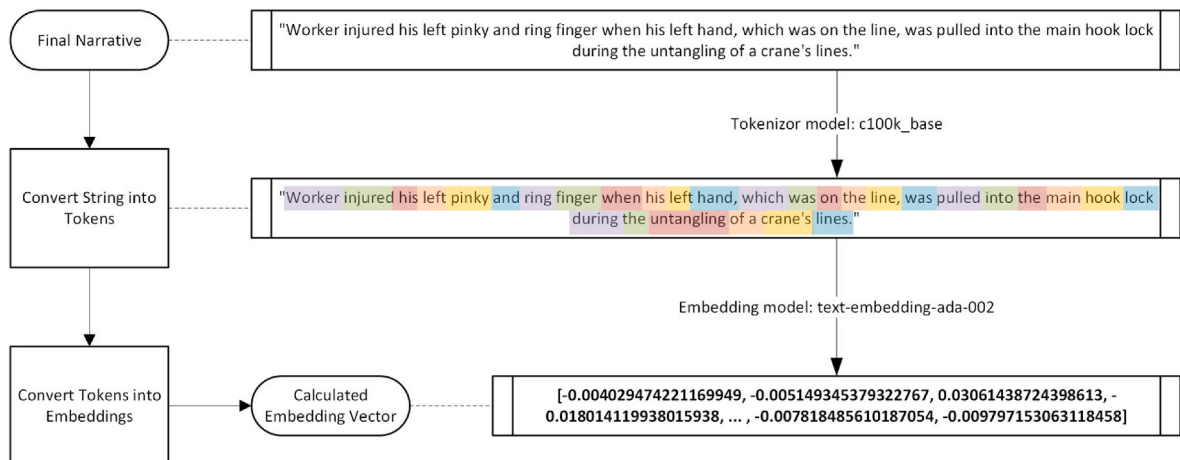


Fig. 2. Flowchart of calculating embeddings from raw text.

statistical methods for reduction. Traditional techniques, such as Principle Component Analysis (PCA), struggle in handling data of high dimensions due to the challenge of preserving distances between nearby datapoints and modeling curved manifolds (van der Maaten and Hinton, 2008), or spaces that exist in other geometries besides flat planes. Hybrid approaches combining PCA and Latent Dirichlet Allocation (LDA) have shown promise in reducing dimensionality and improving related text-mining tasks, indicating the need for alternative methods to address high-dimensional data challenges (Singh et al., 2022).

To overcome this challenge of dimensionality reduction for the visualization of the resulting clusters in two dimensions, t-SNE, or t-Distributed Stochastic Neighbor Embedding, is utilized with the following parameters: n-components (2), perplexity (15), random state (42), and learning rate (200). In conjunction with K-means clustering, t-SNE is a prominent approach to aid in the visualization of high-dimensional spaces (van der Maaten and Hinton, 2008; Treisman et al., 2022). Unlike existing techniques, t-SNE excels at creating a unified map that reveals structures at various scales (in this approach, from 1536 to two dimensions), a crucial feature for high-dimensional data distributed across multiple interconnected low-dimensional spaces (van der Maaten and Hinton, 2008). Since dimensionality reduction is only used to assist in the visual representation of the resulting K-means clusters, the selection of the algorithm is not crucial other than ensuring that clusters appear distinct and unique. Future studies may explore the validation of the associated algorithms and potentially utilize the reduced dimensions in clustering, especially if semantic representation is maintained.

While inherently useful for the nature of the data embedded in this study, the K-means clustering algorithm presents several additional challenges aside from the limitations of visualizing high dimensions. One challenge is the sensitivity to outliers, which can significantly alter the cluster mean value (or centroid) (Usino et al., 2019). The algorithm also has variations in choosing the initial centroids, which can impact the clustering results (Balbi, 2010). Another challenge is the curse of multidimensionality, which has interesting effects on clustering methods, particularly on how distances between points are computed (Balbi, 2010). Additionally, there is no specific solution to find the exact value of  $n$ , or number of clusters, to partition the training dataset, which can make it challenging to determine an appropriate number of clusters (Yassin, 2020).

The high-dimensional nature of OpenAI's embeddings introduces complexity beyond the conventional evaluation of K-means clustering. The elbow method, aided by the  $SSE$  (sum of squared errors) defined in Eq. (4), is commonly utilized to determine an optimal number of clusters (Lossio-Ventura et al., 2021; Ma et al., 2021). The  $SSE$  evaluates each data point ( $x$ ) in relation to its associated cluster centroid ( $\bar{X}$ ) across all  $i$  points in  $j$  total clusters. Apart from maintaining consistency with the proposed approach, other methods such as silhouette score may aid in the validation and optimization of clusters (Weng et al., 2022). Later results in this study indicate that the number of clusters in this type of analysis is not implicitly critical; thus, only the elbow method is employed.

$$SSE = \sum_{i=1}^p \sum_{j=1}^n |x_i - \bar{X}_j|^2 \quad (4)$$

During the application of the elbow method, the number of clusters selected for the K-means algorithm is systematically varied. By plotting the total  $SSE$  for all clusters against the number of clusters ( $n$ ), a kink-point may emerge, indicating the point where the rate of change in error is most pronounced. This inflection point designates a potential optimal number of clusters. However, due to the nature of embeddings, small error propagates over the entire length of the vector which may result in a relatively large error for each cluster, discerning no distinguishable kink-point. This propagated error might be exacerbated if the centroids of each cluster were also not initialized correctly, rendering

many cluster optimization techniques rudimentary.

To address some of these limitations and enhance the effectiveness of the employed clustering method for embeddings, we propose a new approach. In addition to manually disseminating the resulting two-dimensional t-SNE cluster visualizations and employing the elbow method, we introduce a refinement step utilizing representative summary embeddings to describe each cluster centroid. This summarization step is elaborated on in the forthcoming section on prompting. Given that the original cluster centroids are arbitrarily generated by the K-means algorithm, they may not accurately capture the optimal centroid locations. By deriving significant embeddings for these centroids, we aim to improve the accuracy of the resulting clusters, thus providing more representative insights into construction safety incidents.

## 2.5. Large language models

The Transformer, the main architecture behind LLMs, exhibits robust performance in tasks such as machine translation, document generation, and syntactic parsing (Radford et al., 2018). Comprising an encoder, decoder, and attention mechanism, probabilities  $p(x)$  are predicted for symbol sequences ( $s_1, s_2, \dots, s_n$ ) from a set of learned examples ( $x_1, x_2, \dots, x_n$ ) by factoring them into a product of conditional probabilities (Radford et al., 2019). This involves assessing the likelihood of a word occurring based on the words surrounding it in an unsupervised fashion, as depicted in Eq. (5). When presented with input data, the model utilizes this probabilistic framework to generate coherent and contextually appropriate outputs. For instance, given a safety incident description, the model can produce a detailed safety report by selecting words and phrases that align with the provided context and adhere to the syntax of the language.

$$p(x) = \prod_{i=1}^n p(s_i | s_1, \dots, s_{n-1}) \quad (5)$$

The landscape of NLP has been revolutionized by such language models, and increasing the size of training data has demonstrated various advantages, including enhanced performance and increased sample efficiency (Wei et al., 2023). In various industries, including construction, LLMs hold the potential to improve efficiency, accuracy, and communication (Prieto et al., 2023). This study utilizes OpenAI's GPT-3.5 (version: gpt-3.5-turbo-1106) for cluster summary curation and incident classification based on text alone. While newer models with similar or improved performance have since emerged, GPT-3.5's selection is crucial, given its status as one of OpenAI's largest and most performant.

## 2.6. LLM prompting

Interfacing requires utilizing predefined templates referred to as a prompt to guide the generation process for producing task-specific responses (Kojima et al., 2023). These methods support innovative techniques such as few-shot learning, in-context learning, and chain-of-thought prompting (Wei et al., 2022; Kojima et al., 2023). Zhou et al. emphasize the importance of careful prompt engineering, proposing a method to optimize instructions directly within the natural language hypothesis space using LLMs (Zhou et al., 2023).

Recent models like GPT-2 (and proceeding iterations), showcase notable zero-shot and few-shot capabilities, excelling in diverse NLP tasks and achieving state-of-the-art performance on numerous datasets (Radford et al., 2019). These capabilities leverage transfer learning, enabling the execution of tasks without examples (zero-shot) or with examples during prompt inferencing (few-shot) (Brown et al., 2020). All prompting in this study is carried out through OpenAI's Application Programming Interface (API), which allows GPT-3.5 to be used through proprietary web servers. The parameters for each inference are as follows: temperature (0.1), max tokens (4,000), max retries (6), and top-p

(0.4). Python was primarily used to facilitate data processing, prompt selection/injection, and figure curation.

### 2.6.1. Summarization

While the K-means clustering of the embeddings produces intricate similarities and differences between the associated incidents, aside from manual dissemination it is difficult to navigate a large dataset and derive useful information that can be communicated to the field. Thus, after the optimal number of clusters is determined, GPT-3.5 is prompted to summarize each cluster. Fig. 3 details the methodology used to summarize individual clusters. Since the LLM has a limited context window, or amount of text that can be processed in one inference, the input must be chunked accordingly.

The MapReduce algorithm (Dean and Ghemawat, 2008), Eq. (6), is employed to process the dataset of road construction incidents, each associated with the accident narrative, across all clusters denoted as  $M$ . Within each  $m^{\text{th}}$  cluster, incidents are further divided into  $N$  chunks based on token lengths to fit within the 4096 token context window of gpt-3.5-turbo. The map phase independently processes each chunk, generating individual  $n$  chunk summaries for cluster  $m$  ( $C_{m,n}$ ). The reduce phase then consolidates these chunk summaries for each cluster  $m$  into a final summary  $S_m$ , achieved through a reduction operation across all chunks within the cluster. This cluster-specific summarization is repeated  $M$  times, providing comprehensive and aggregated insights into the road construction incidents across all distinct clusters.

$$S_m = \text{Reduce}\{\text{Map}(Chunk_{m,1}, Chunk_{m,2}, \dots, Chunk_{m,N})\} \quad (6)$$

This process involves two distinct prompts to generate comprehensive summaries for individual clusters. Given that many clusters may contain numerous incidents surpassing the context window, the initial “Map Prompt” acts as an abstraction layer. For instance, if there are 10 incident descriptions provided, a summary is created solely for these incidents. To produce a conclusive summary representing a broader cluster of accidents, the concluding “Reduce Prompt” is utilized. Here, the summarized chunks from the mapping function are consolidated into one cohesive summary.

A final self-consistency check was implemented to reduce ancillary or hallucinated information from intermediate “Reduce” summaries, an adaption of the SelfCheckGPT approach from (Manakul et al., 2023), designed to enhance the reliability of Generative LLMs. Unlike existing fact-checking methods, this approach operates in a zero-resource manner, meaning it doesn't rely on external databases. Instead, it utilizes a sampling-based strategy, leveraging the notion that accurate information yields consistent responses, while hallucinated facts result in divergent and contradictory answers. In this study, the Map phase intentionally prompts the model to produce a list of facts ( $f_j$ ) for each group of incidents in  $N$  chunks, producing a total list of facts for cluster  $m$  ( $f_{m,j} = f_{m,1}, f_{m,2}, \dots, f_{m,J}$ ). Once a consolidated summary from the Reduce phase is produced, the LLM is then prompted to evaluate if fact  $j$  aligns with  $S_m$  for all  $J$  facts:

'''

Summary: {summary,  $S_m$ }

Fact: {fact,  $f_{m,j}$ }

Does this fact support the summary above?

Answer “Yes,” “No,” or “N/A”:

'''

Eq. (7) then computes an inconsistency score ( $IS$ , %) for facts and summaries in each cluster  $m$ . For each comparison between fact  $j$  and summary  $S_m$ , the indicator variable  $\lambda_j$  maps a score into three conditions: high alignment true assertion (“Yes,”  $\lambda_j = 0.0$ ); negative alignment false assertion (“No,”  $\lambda_j = 1.0$ ); and minor alignment inconclusive assertion (“N/A,”  $\lambda_j = 0.5$ ). A higher score correlates to many facts being divergent from the original summary, implying that the model may have hallucinated during generation or some facts were inconsistent with the main theme of the cluster. A final, high alignment summary  $S_m^*$  is then crafted for each cluster using only true assertion facts, fostering a more reliable and focused description.

$$IS_m = 100 \cdot \frac{1}{J} \sum_{j=1}^J \lambda_j |f_{m,j}, S_m| \quad (7)$$

### 2.6.2. Classification

Following cluster summarization, the LLM is employed to classify incidents individually. This classification not only provides insights on the reliability of the original categorical entries, but also gauges the overall quality of the abstracts. Deviating slightly from traditional prompting techniques, two methods are assessed: zero-shot and few-shot. These prompts, carefully curated for performance, guide the model to produce output for evaluation. The categorical entries considered for classification, as discussed in the previous section outlining the IMIS database (Table 1), include “event\_type”, “nature\_of\_inj”, “part\_of\_body”, “evn\_factor”, “hum\_factor”, “occ\_code”, “src\_of\_injury”, and “degree\_of\_injury”.

The zero-shot prompting, or null example, technique provides the model with a list of unique entries from the original database and the associated abstracts. Without considering the original classifications, the model is asked to classify the specified categorical variables within the context of the abstracts using only the provided unique values. A well-written abstract will correspond to the original coding for each categorical variable if it contains enough contextual relevance, justifying the human's original selection. For example, if a narrative explicitly describes a body part involved in an accident, it should match the original coding for the “part\_of\_body” column in the database. Any disagreement between the prediction and the human classification may suggest that the narrative alone lacks sufficient information to derive a specific classification, such as the type of incident or its contributing factors.

The few-shot prompting technique reassesses the original coding in the specified categorical variables, within the context of the main abstracts. Here, the original classification is presented in the prompt, along with a list of all unique entries for the specified category, enabling the LLM to propose a more representative classification based on the full narrative. The term “few” in few-shot refers to the limited number of examples or shots needed for the model to learn effectively. For this study, a three-shot configuration was implemented per classification category. For each instance, three classification examples were provided at random from a predefined dictionary. While increasing the number of shots may generally improve performance, the arbitrary selected value of three in this investigation provides a balance between accuracy and token-efficiency – as more examples would increase the compute required per inference.

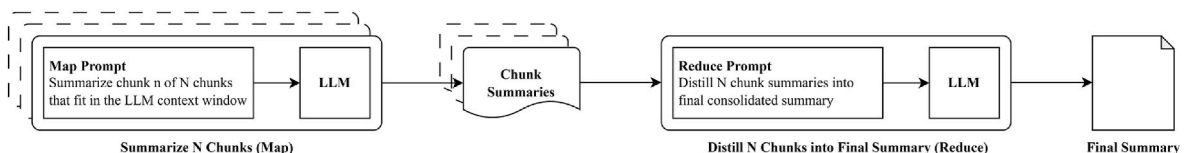


Fig. 3. MapReduce prompt template for cluster summarization.

This approach leverages the language model's ability to learn by example, commonly referred to as in-context learning, enabling it to make more informed decisions based on the original entries. While the prediction may often agree with the original classification, a deviation could indicate a potentially more suitable classification based on the information provided in the abstract. Alternatively, regardless of the applied classification technique, the LLM may misinterpret the user's instructions, a phenomenon commonly referred to as "hallucination." This misinterpretation could lead to additional discrepancies between the original coding and the predicted classification, beyond those previously mentioned. In the zero-shot prompting technique, where the model is not provided with examples, these discrepancies may be more pronounced compared to the few-shot method, where the model receives a sufficient number of examples. Moreover, if the LLM fails to derive a more appropriate classification in the few-shot scenario, it may default to the original human entry, an option unavailable in the zero-shot approach.

Performance metrics typical to multi-class and binary classification, Eqs. (8)–(11), are then applied to the resulting classifications to assess overall performance with both prompting methods (Ma et al., 2021; Yassin, 2020). *TP* (True Positives), *TN* (True Negatives), *FP* (False Positives), and *FN* (False Negatives) are used to help quantify accuracy, recall, precision, and the F1 Score of the model based on its predictions compared to the original classifications.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (8)$$

$$Recall = \frac{TP}{TP + FN} \quad (9)$$

$$Precision = \frac{TP}{TP + FP} \quad (10)$$

$$F1\ Score = 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)} \quad (11)$$

The accuracy metric gauges the general correctness of the classifications, while recall and precision assess its capability to identify positive predictions or match the original coding in the database. The F1 Score provides a balanced evaluation by considering both false positives (*FP*) and false negatives (*FN*), harmonizing precision and recall metrics.

### 3. Results and discussion

#### 3.1. Cluster analysis

The vectors generated by OpenAI's embedding model were initially computed and cached for individual abstracts sourced from the IMIS

database. Upon establishing this foundation, the K-means algorithm was applied, which entails a multitude of initialization hyperparameters. These parameters, ranging from the number of maximum iterations to centroid sampling and seeding, remained constant throughout this study. Fig. 4 shows the relationship between total *SSE* (left) and the average cluster-wise *SSE* (right) versus the number of clusters. This figure reveals the persistent relatively large error as the number of clusters increases. While more clusters generally lead to a reduction in error, particularly noticeable in the average cluster-wise error, this reduction alone is not significant enough to determine optimal groupings of incidents.

From this point, determining the optimal number of clusters is still not trivial in this case, as there was no distinct kink-point with the elbow method (Fig. 4). It is evident that small errors in comparing individual data points to cluster centroids, through Euclidean distance, gets propagated through the high dimensions of the resulting embedding vectors, attributing to the curse of dimensionality. Given the large number of parameters controlling the unsupervised K-means algorithm and the lack of inflection points from the resulting *SSE* analysis, the use of different initialization parameters may enhance these results.

To address this propagated error, the selection of the optimal number of clusters (*n*) via reduction of *SSE* was coupled with a manual review of resulting clusters and their corresponding incidents. Fig. 5 demonstrates a sample of varying numbers of clusters, specifically showing three (Fig. 5a), ten (Fig. 5b), and fourteen (Fig. 5c) clusters reduced from 1536 dimensions to two dimensions using the t-SNE algorithm for visualization. In each figure, the left plot depicts the clustering for the selected number of accident categories, accompanied by a distribution plot on the right indicating the number of incidents in each cluster. The color and arbitrary shape assigned to each cluster serve solely to differentiate between clusters.

These specific numbers of clusters (three, ten, and fourteen) were chosen to provide a diverse representation of the clustering results and to demonstrate how the distribution of incidents varies across different cluster counts. Three clusters exhibited a relatively high total *SSE* of 132.7, which was the first instance that the error was significantly reduced from one or two clusters; however, individual cluster-wise *SSE* averages 44, which explains the less centric organization of incidents. The total *SSE* does not drastically decrease from three to fourteen clusters as the number of clusters increases. Instead, the decline in total error and average cluster-wise error gradually decreases as the number of clusters increases.

Relying solely on error for determining the optimal number of clusters may be a rudimentary approach, especially in high-dimensional spaces. In such scenarios, it becomes challenging to find an optimal number of clusters based solely on error metrics, as the *SSE* may not sufficiently capture the underlying structure of the data. Other methods,

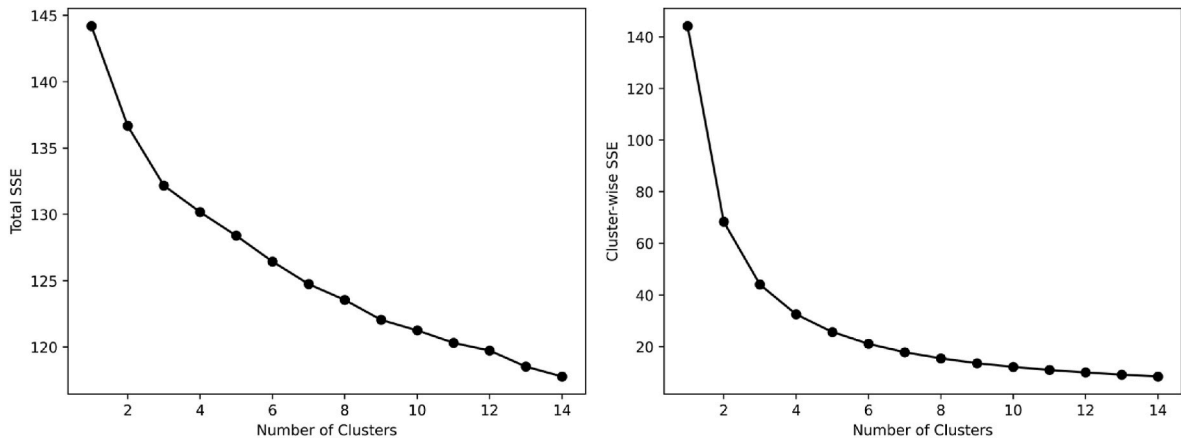


Fig. 4. Total (left) and average cluster-wise (right) *SSE* versus number of clusters.

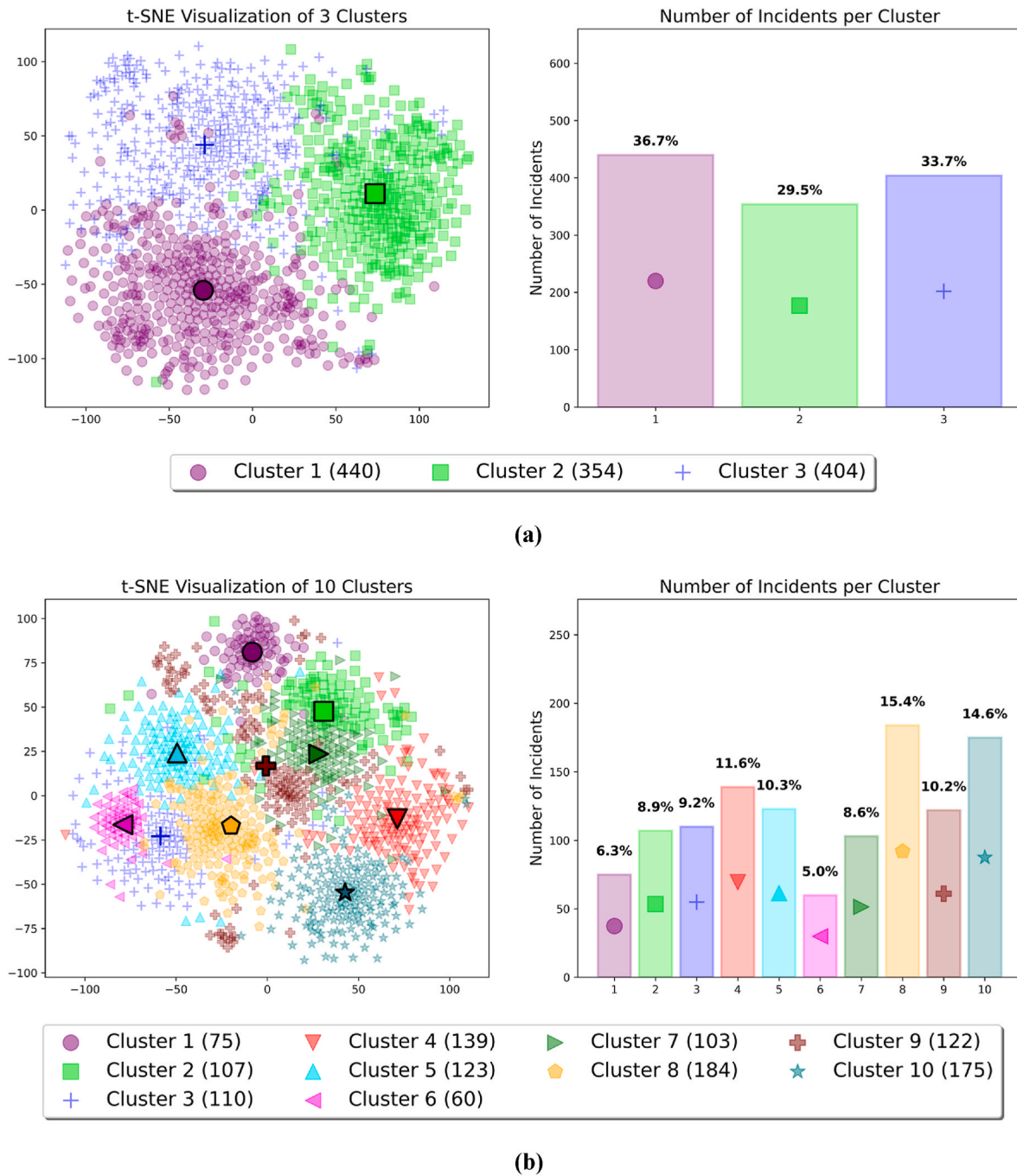


Fig. 5. (a) Three, (b) ten, and (c) fourteen clusters of road construction injuries with t-SNE visualization.

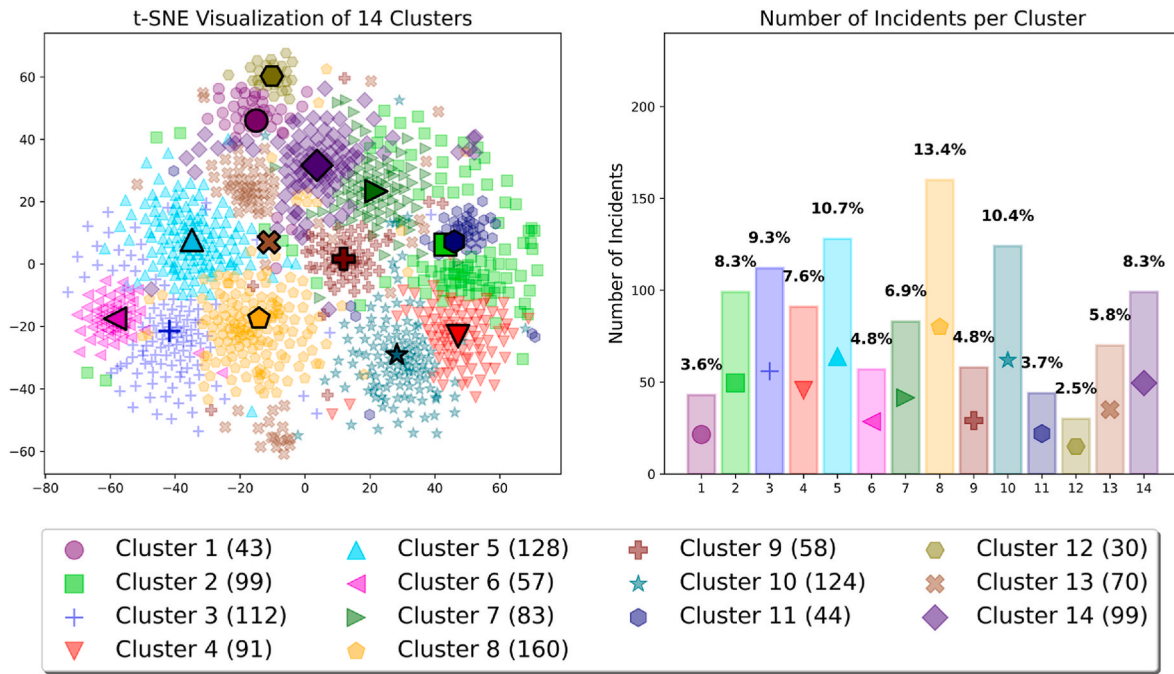
such as silhouette analysis, may provide additional insights into the appropriate number of clusters by considering both within-cluster homogeneity and between-cluster separation.

Ultimately, ten clusters (Fig. 5b) were selected for further analysis to demonstrate the utility of this approach. As the number of clusters increases, overarching accident types diffuse into smaller, more specific subcategories. Purely from a visual perspective (after dimensional reduction), the ten selected clusters remain distinct and centric, which is preferred in a Euclidean distance-based algorithm. Clusters 4, 8, and 10 in provide the highest contributions to overall SSE, insisting that there may be prominent outliers. In comparison, although some of the fourteen clusters presented in Fig. 5c appear more centric, the total SSE does not significantly decrease from the optimal ten selected. The following

summarization of ten clusters and additional cluster refinement justifies the optimal number of clusters, aside from only analyzing error with manual visualization.

### 3.1.1. Cluster summaries

Summaries of the ten clusters were then generated using the Map-Reduce template provided in Fig. 6. The prompt was specifically engineered to request a list of facts rather than a narrative summary to maximize information density and minimize the linguistic filler that can lead to model hallucinations, aligning directly with the required format for the subsequent self-consistency evaluation. Since the context window for GPT-3.5 was limited to 4096 tokens, the map prompt sufficiently summarizes varying token-limited chunks for each cluster. For



(c)

Fig. 5. (continued).

| Map Prompt                                                                                                                                                                                                                                                                                                                                 |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Given the following road construction incidents, extract a list of {n_samples} facts that are common between them. Try to be specific and identify unique relationships between the incidents, rather than being generic about construction incidents.                                                                                     |    |
| Format your output as a bulleted list.                                                                                                                                                                                                                                                                                                     |    |
| Incidents:                                                                                                                                                                                                                                                                                                                                 | \\ |
| {text}                                                                                                                                                                                                                                                                                                                                     | \\ |
| Reduce Prompt                                                                                                                                                                                                                                                                                                                              |    |
| The following is a list of facts about a specific group of road construction incidents:                                                                                                                                                                                                                                                    |    |
| Facts:                                                                                                                                                                                                                                                                                                                                     | \\ |
| {text}                                                                                                                                                                                                                                                                                                                                     | \\ |
| Take these and distill them into a final, consolidated summary paragraph of what the road construction incidents have in common. Try to be specific and identify unique relationships between the incidents, rather than being generic about general construction safety or providing details that pertain to only one or a few incidents. |    |

Fig. 6. MapReduce summarization prompt template for the GPT-3.5 LLM.

example, cluster 1 was partitioned into eight chunks and distilled into one conclusive summary for its associated cluster. This approach ensured that no individual cases were excluded from summarization, procuring the most representative summary for the corresponding clusters. Specific instructions such as “try to be specific” and “identify unique relationships” embedded in the prompt guided the model to generate more meaningful responses, and altering the prompt can steer

the response in the wrong direction. The LLM was prompted to generate fifteen facts ( $\{n\_samples\}$ ) for each chunk.

Self-consistency results from MapReduce summarization, including inconsistency scores, and the number of true and false assertions, are displayed in (Table 2). These findings indicate that the model was consistent in 83% (1551 of 1860) of generated facts, where the remaining 17% were a product of either hallucination or noisy incident reports. The least consistent group was cluster 8, resulting in 69 false and 71 inconclusive assertions, respectively. Moderate consistency was presented in clusters 5–7, 9, and 10. In contrast, facts generated for clusters 1–4 were highly consistent with their intermediate summaries.

The resulting summaries ( $S_m^*$ ) are presented in Table 3, which is a product of revising the summaries from the reduce stage ( $S_m$ ) in the provided prompt template (Fig. 6) by omitting any false or inconclusive facts. Intermediate chunk summaries from the map stage are not provided in this paper for spatial limitations. Within Table 3, the first column signifies the distinct cluster number corresponding to the previous t-SNE visualized plots, the number of incidents compared to the total, alongside the associated percentage of the total. In the second column, the generated summaries remain unperturbed from the raw output. To

Table 2

Self-consistency of generated cluster summaries from MapReduce.

| Cluster | No. Chunks (N) | No. Facts (J) | True Asser. | False Asser. | Inconclusive | IS (%) |
|---------|----------------|---------------|-------------|--------------|--------------|--------|
| 1       | 8              | 120           | 115         | 3            | 2            | 3.33   |
| 2       | 11             | 165           | 165         | 0            | 0            | 0.00   |
| 3       | 11             | 165           | 165         | 0            | 0            | 0.00   |
| 4       | 14             | 210           | 207         | 0            | 3            | 0.71   |
| 5       | 13             | 195           | 153         | 20           | 22           | 15.9   |
| 6       | 6              | 90            | 71          | 5            | 14           | 13.3   |
| 7       | 11             | 165           | 143         | 6            | 16           | 8.48   |
| 8       | 19             | 285           | 144         | 69           | 71           | 36.6   |
| 9       | 13             | 195           | 160         | 4            | 31           | 10.0   |
| 10      | 18             | 270           | 228         | 7            | 35           | 9.07   |
| Total   | 124            | 1860          | 1551        | 114          | 194          | ~      |

**Table 3**  
Resulting LLM generated summaries for ten clusters.

| Cluster                             | Generated Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cluster 1<br>75/1198<br>6.3 %       | The road construction incidents share several common factors. Firstly, the incidents involved employees operating heavy machinery or equipment such as rollers and compactors. Secondly, the accidents often resulted in severe injuries or fatalities, with employees being crushed, pinned, or run over by the machinery. Thirdly, multiple incidents occurred when the employees lost control of the equipment, leading to accidents such as rollovers or collisions with obstacles. Additionally, seatbelt usage and the presence of safety equipment like rollover protection structures were inconsistent, contributing to the severity of the injuries. The incidents occurred in various locations and at different times, involving employees from government agencies or construction companies. Overall, these incidents highlight the importance of proper training, equipment maintenance, and adherence to safety protocols in road construction.                                                                                                                                                                                                                                                                                                                                                                                         |
| Cluster 2<br>107/<br>1198<br>8.9 %  | The road construction incidents share several common factors. Firstly, employees were frequently injured or killed while working on road construction sites, often due to being struck or run over by construction vehicles or equipment. Lack of proper visibility, unsafe conditions, and failure to use safety measures were contributing factors in many incidents. Heavy machinery, such as loaders, graders, and backhoes, were involved in multiple incidents, with employees being injured or killed by the equipment they were operating. Seat belt non-usage and operable safety device failures were also identified as contributing factors. The incidents occurred over a span of several years and in various locations, with employees sustaining a range of injuries including fractures, crushed limbs, and head injuries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cluster 3<br>110/<br>1198<br>9.2 %  | The road construction incidents involved multiple employees being struck by vehicles while working on various road construction sites. The incidents occurred at different times of the day and in different weather conditions, including heavy fog and low visibility. Trucks and tractor trailers were involved in several incidents, and some employees were struck by vehicles that veered off the road or crossed the center line. Inadequate warning signs or barricades were present in some incidents, and in one case, a construction dump truck's backup alarm was not operational. The incidents occurred on different types of roads, including highways, freeways, and airport runways. Multiple employees were involved in some incidents, with varying degrees of injuries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cluster 4<br>139/<br>1198<br>11.6 % | The road construction incidents between 2004 and 2009 shared common factors that contributed to accidents and injuries. These incidents frequently involved cranes and aerial lifts, with incidents ranging from tipping over to boom collapses and contact with power lines. Falling objects, such as rebar and concrete beams, also caused injuries. Unstable surfaces, poor welding, and equipment malfunctions were contributing factors in several incidents. Lack of proper training, safety measures, and communication were evident in some cases. Multiple incidents involved employees being electrocuted due to contact with power lines, highlighting the importance of proper training and knowledge of safe operating procedures. Improper rigging or securing of loads was a recurring issue, leading to employees being pinned or crushed by heavy objects. Inadequate maintenance and inspections, as well as failure to follow substance abuse policies, were observed in some incidents. The use of recommended equipment and adherence to manufacturer's load charts were not consistently followed, resulting in accidents. These incidents emphasize the need for comprehensive safety programs, including training, proper equipment use, and regular inspections to prevent accidents and protect workers in road construction. |
| Cluster 5<br>123/<br>1198<br>10.3 % | The road construction incidents involved workers being struck or run over by vehicles, with dump trucks being the most common type of vehicle involved. In several incidents, workers were struck and killed by dump trucks that were backing up. Lack of visibility due to clothing or obstructions was a potential causal factor in some incidents. The incidents occurred at different times of the day and on different dates, spanning from 2000 to 2011. The workers were engaged in specific tasks related to road construction, such as filling potholes or grinding asphalt. Multiple incidents resulted in fatalities, highlighting the serious risk to employee safety. Communication and visibility were identified as important factors in preventing these incidents.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Cluster 6<br>60/1198<br>5.0 %       | The road construction incidents involved flagmen or flaggers who were struck by vehicles while directing traffic. The incidents occurred in various locations and resulted in severe injuries or death for the workers involved. Vehicles such as tractor trailers, milk tanker trucks, cars, and pickup trucks were involved in the incidents, traveling at                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**Table 3 (continued)**

| Cluster                              | Generated Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cluster 7<br>103/<br>1198<br>8.6 %   | speeds ranging from approximately 35 to 60 miles per hour. In some cases, the workers were struck by equipment on the vehicles before being run over by their rear wheels. Factors such as sun glare, failure to obey directions, and impaired vision contributed to the incidents. Warning signs, cones, and high visibility clothing were present at the construction sites but did not always prevent the accidents. Overall, these incidents highlight the importance of proper training, adherence to safety protocols, and increased awareness from both drivers and workers in road construction zones.                                                                                                                                                                                                                                                                                                                                                                 |
| Cluster 8<br>184/<br>1198<br>15.4 %  | The road construction incidents that occurred between July 3, 2003, and April 24, 2013, share several common factors. All incidents involved road construction activities and resulted in injuries or fatalities to the workers involved. Excavators were frequently implicated in the incidents, with employees being caught under, run over, or struck by the equipment. Falling objects, such as rocks or concrete, were also a common hazard. Trench collapses, equipment malfunctions, and unstable soil contributed to the accidents. Fractures, crush injuries, and suffocation were among the types of injuries sustained. Proper safety measures, training, and equipment support are crucial in mitigating the risks associated with road construction work.                                                                                                                                                                                                         |
| Cluster 9<br>122/<br>1198<br>10.2 %  | The road construction incidents share several common factors. Multiple incidents involved employees being struck by vehicles, including dump trucks, cars, and trains, resulting in fatalities. Some incidents occurred during late dusk to dark hours, highlighting the potential dangers of low visibility. In some incidents, employees were not wearing reflective vests, which may have contributed to the accidents. The incidents occurred during various road construction activities, such as grading, paving, and sign installation. Lack of proper safety measures, such as backing up without a functioning alarm or not using a ground guide, were contributing factors in some incidents. These incidents emphasize the importance of implementing safety protocols and precautions to prevent fatalities and injuries in road construction zones.                                                                                                               |
| Cluster 10<br>175/<br>1198<br>14.6 % | The road construction incidents involved employees working in various locations and activities, such as bridge construction projects, city streets, highways, and excavation sites. Injuries were sustained in all incidents, with multiple employees being hospitalized. The incidents highlighted the importance of proper safety procedures and equipment in road construction. Common types of injuries included amputations, burns, fractures, and lacerations, often caused by machinery or equipment. Power tools, such as grinders and saws, were frequently involved. Some incidents were attributed to equipment malfunctions or failure to follow safety procedures. Employee #1 was involved in multiple incidents, experiencing various injuries and hospitalizations. Heat-related issues and dehydration were also reported in some incidents. Overall, these incidents emphasize the need for comprehensive safety measures and training in road construction. |

demonstrate distinction between true, false, and inconclusive fact assertions, the following are samples from cluster 6, which primary centers around flaggers involved in struck-by hazards:

- **True Assertions:** “Multiple incidents involved flaggers who were struck by vehicles while directing traffic,” “Some incidents involved multiple vehicles, such as the truck striking a stopped vehicle before hitting the worker,” “Reflective vests were worn by employees in some incidents, but they did not provide sufficient protection,” etc.
- **False Assertions:** “The incidents occurred on various dates between 2004 and 2006,” “In one incident, the employee received several bullet impacts before being transported to the hospital and dying a few hours later,” etc.
- **Inconclusive:** “In several incidents, the drivers who struck the flagmen did not slow down and fled the scene,” “A construction laborer fell from a moving trailer and struck his head on the pavement, resulting in his death,” etc.

While a majority of facts were supportive, some do not align with major commonalities within the overarching theme; thus, withheld from the final revised summary generation.

Since 52% of cases were attributed to “struck-by” in the “event\_type” column of the original database, as shown in Table 1, these types of accidents prevailed in multiple cluster summaries (Table 3). Initially, before instructing the model to explain unique relationships between the clusters, general observations related to incidents involving struck-by accidents were consistently reappearing. After the model was guided to be more specific, subtle differences in the type of struck-by accidents surfaced. For example, cluster 2 specifies accidents involving struck-by construction vehicles or equipment and cluster 7 alludes to a specific type of equipment (excavators). Cluster 3 generally refers to incidents involving workers being struck by third-party vehicles, vehicles entering the work zone, etc. The generated summary for cluster 8 persists as a very general description of the commonalities between the incidents, potentially alluding to incidents that do not belong to other well-defined clusters.

Alternatively, many other clusters remain distinct. Cluster 1 pertains to machinery accidents, often involving crushed or pinned injuries. The incidents in cluster 4 typically involve workers subjected to elevated risks, such as working from cranes or lifts. Bodily injuries and traumas, often as a result of machinery misuse, were clustered together in cluster 9. Finally, cluster 10 alludes to bridge construction and its associated activities.

### 3.1.2. Cluster refinement

Since the incidents are clustered semantically, or based on their calculated embeddings, taking advantage of this relationship allows for a more refined cluster initialization. Traditionally, the K-means algorithm remains unsupervised, which is attractive to many researchers for its ease of use and relatively low compute. In this case, since clusters are now summarized based on distinct commonalities, the algorithm is re-initialized with calculated embeddings of the generated summaries as centroids (instead of randomized). In the event that an incident changed its position from the original cluster to another cluster in this refinement, this may have indicated improper or poor initial clustering. Fig. 7 illustrates the clustering refined with the generated summaries as centroids.

A comparison of Figs. 5 and 7 reveals that cluster 8 exhibited poor initial clustering, as it essentially vanished; therefore, it is likely that 9 clusters more closely resemble the major types of accidents instead of 10. Alternatively, clusters 1, 4, 9, and 10 do not present significant changes. Through an extension of manual interpretation of the generated summaries and their associated abstracts in the original database, we were able to identify commonalities within each cluster. Cluster 1 still generally pertains to machinery and equipment accidents (typically involving crushed or pinned injuries). Cluster 4 represents incidents frequently involving cranes and aerial lifts (including falling objects, unstable surfaces, etc.). Cluster 9 then also includes machinery and equipment accidents, but in this case represents body injuries such as fractures, burns, amputations, etc. Finally, cluster 10 encapsulates accidents that primarily involve employees working on bridges (mainly consisting of falls, falsework failures, etc.).

This comparison also indicates that clusters 2,3, and 5 through 7 significantly changed from their original grouping. Cluster 2 consists of struck by accidents where employees were run over by heavy equipment (highlighting lack of visibility, inadequate seatbelt usage, and other safety device failures). Cluster 3 involves accidents where employees were struck by trucks, tractor-trailers, and other vehicles that veered off of the roadway (with inadequate warning signs or barricades). Cluster 5 pertains to struck-by accidents that typically involved dump trucks, several of which were where the dump truck was backing up. Cluster 6 is usually associated with flaggers who were struck by vehicles while directing traffic. Finally, cluster 7 encapsulates struck by accidents involving excavators and/or falling objects. All of these summaries have

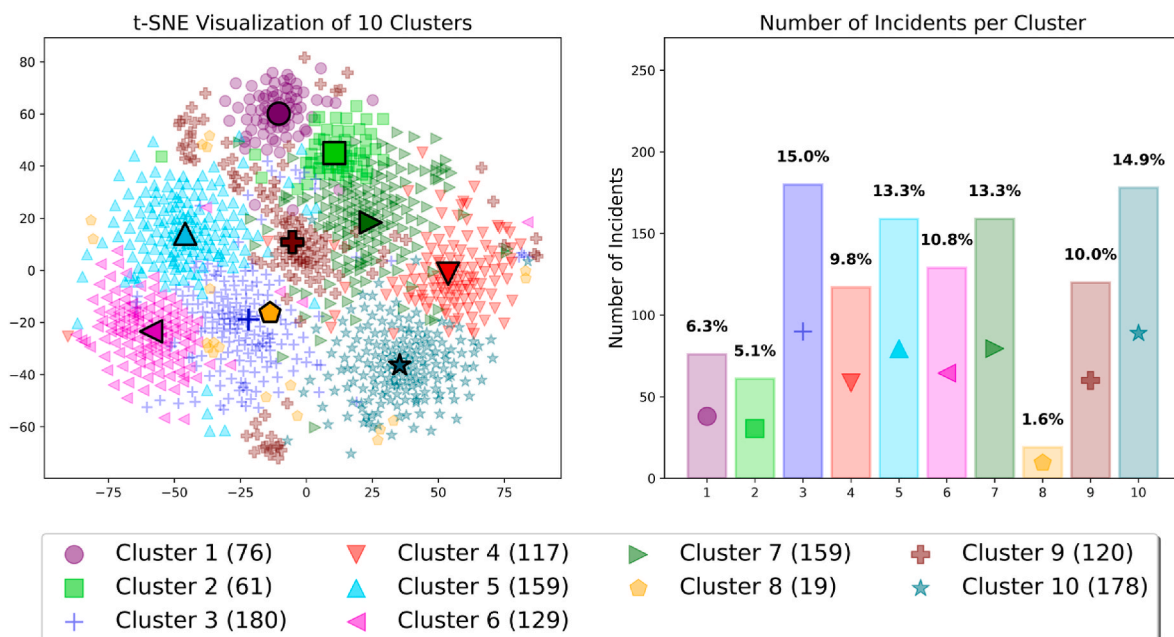


Fig. 7. Ten clusters initialized with embeddings of LLM generated summaries.

some correlation to struck-by accidents, as discussed previously, and constitute approximately 57.5% of all cases.

### 3.1.3. Relationship to original IMIS database

To conveniently display the resulting cluster summarization, the following titles were manually curated for each cluster based on the generated text, coupled with their associated distribution between the ten clusters.

- Cluster 1. Machinery Incidents: Crushed and Pinned Injuries (6.3%)
- Cluster 2. Poor-visibility Hazards: Heavy Equipment Run-overs (5.1%)
- Cluster 3. Roadside Risks: Vehicle Strikes Beyond Barriers (15.0%)
- Cluster 4. Fall Incidents: Crane and Aerial Lift Incidents (9.8%)
- Cluster 5. Dump Truck Operation: Backing-Up Struck-by Accidents (13.3%)
- Cluster 6. Traffic Control Risks: Flaggers Hit While Directing (10.8%)
- Cluster 7. Excavator Operation: Struck-by and Falling Objects (13.3%)
- Cluster 8. Inconclusive, Other (1.6%)
- Cluster 9. Machinery Risks: Body Injuries and Trauma (10.0%)
- Cluster 10. Bridge Work Risks: Falls and Falsework Failures (14.9%)

The relationship between the derived categories to those presented in the original database is not trivial, which makes validation of the approach difficult. The IMIS provides highly aggregated breakdowns of individual accidents into multiple columns. This format makes it challenging to procure meaningful or actionable insights for researchers or practitioners. As mentioned in the description of the database (Table 1), for example, “event\_type” comprises 14 unique entries while “src\_of\_injury” has 43. The distillation of these two fields alone creates far too many subcategories to disseminate. Additionally, the combination of these two columns only represents a fraction of the aggregated subcategories from the database.

To demonstrate this phenomenon, the “Struck By” and “Struck Against” entries in the “event\_type” column of the database can be grouped due to similarity and lack of supplemental description. Together, these entries make up a combined 55.4% of reported accidents. When coupled with the “src\_of\_injury” column, 27 of the 43 unique entries prevail, with the top five presented below along with their percentage of total accidents.

- Struck By/Struck Against: Motor Vehicle – Highway (23.0%)
- Struck By/Struck Against: Motor Vehicle – Industrial (8.5%)
- Struck By/Struck Against: Materials Handling Equipment (5.8%)
- Struck By/Struck Against: Machine (4.3%)
- Struck By/Struck Against: Other (3.9%)

In contrast, the clustering and summarization technique used in this study naturally considers a combination of all factors of an accident as included in each narrative. This approach relies on high-quality narratives that encompass the pertinent details that would allow for IMIS coding during the original incident entry. Following the example of struck-by accidents, clusters 2,3, and 5-7 from the analysis make up 57.5% of reported accidents. Instead of relying on the general subcategories derived from aggregated columns in the database, the summarization defines more specific and inclusive categories such as “Traffic Control Risks: Flaggers Hit While Directing (10.8%)” as exemplified in cluster 6, or “Dump Truck Operation: Backing-Up Struck-by Accidents (13.3%)” from cluster 5. Each of these clusters provides more allusive details including the type of incident, potential sources, worker occupation, etc.

This type of dissection can be crafted for any combination of columns from the original database and eventually compared to the resulting clusters for further validation. The example shown above delineates struck-by accidents, comprising 55.4% of reported accidents in the IMIS

and 57.4% after analysis. However, using only the original database entries fails to convey specific types of equipment used, suffering generality in terms like “Materials Handling Equipment”, rather than the “Dump Trucks” identified post summarization.

Similar results can be shown for other types of accidents. For instance, when combining “Fall (From Elevation)” and “Fall (Same Level)” in the “event\_type” column, fall accidents accounted for 13.9% of incidents in the IMIS and 24.6% after analysis. When coupled with “src\_of\_injury”, only general terms such as “Working Surface” prevail. Alternatively, the analysis surfaces “Cranes and Aerial Lifts” or “Falsework Failures” as supplemental information.

To limit the scope of this study, a full validation of each cluster and its relationship to entries presented in the IMIS database is withheld.

### 3.2. Classification

Similar to the summary generation through curated prompting, the template illustrated in Fig. 8 demonstrates the prompt procured for classification, few-shot for example. The prompt template for zero-shot classification is excluded from the results for its mere similarity to the few-shot prompt. The only notable difference between the two classification prompts is the omission of the prior coding when the model is fed each abstract (zero-shot). The subsequent subsections delve into the immediate interpretation of the resulting classifications, followed by a summary of the overall performance of both techniques.

From an output consistency standpoint, both modes performed exceptionally well. Potential artifacts could have included duplicate/missing entries or hallucinated category selections over the eight classification tasks (e.g., “degree\_of\_inj,” “nature\_of\_inj”). Across all tasks (1198 narratives, eight tasks, two modes) only 48 duplicates were created with zero narrative ID’s missing in the final results. Zero-shot resulted in 263 mislabels (2.7%), while few-shot only resulted in 51 (0.5%). Together, only 1.6% of possible classifications were mislabeled, hallucinating a new label instead of the prescribed unique entries in the prompt – these were treated as “nonclassifiable” for evaluation purposes.

#### 3.2.1. Zero-shot results

In zero-shot classification, the LLM loses the ability to re-evaluate the input, resulting in poorer performance. The model was only provided a list of potential entries to choose from, leading to numerous

#### Classification (few-shot) Prompt

The following incidents have an ID, a(n) {field}, and a description of the incident. The {field} was selected by a human expert based on the following choices:

{unique entries}

For each of the following incidents, please select the most applicable {field} from the list of choices. If another {field} better classifies the incident then what was selected by the human expert, select that instead. If none of the choices are appropriate, please select “Nonclassifiable”, or come up with a brief {field} that is not in the list of choices.

The output needs to be a list of each incident with the original ID and the selected {field}.

For example:

65; “Nonclassifiable”;  
42; “Nonclassifiable”;

Preserve the example format with the existing incident IDs for the output.

Fig. 8. Example classification (few-shot) prompt template for the GPT-3.5 LLM.

“nonclassifiable” entries, or disconnected predictions than what was previously coded. This may suggest a need for more categorical variables per field or improved narrative descriptions. With this approach, two of the three incidents exemplified in Fig. 9 further emphasize these disconnections. In incident #9, the model could not interpret what part of the body was affected by the flagger getting struck by a tractor-trailer, as the abstract did not allude to any specifics. Comparatively, there is no indication of what type of occupational party the employee belonged to in incident #70, resulting in a more general category of “construction laborers”. The remaining example in Fig. 9, incident #340, shows where the narrative contained enough information to classify an accident as “fatal” matching its original designation. Here the narrative explicitly mentions that an employee was crushed and killed during an operation.

Fig. 10 shows the frequently occurring “nonclassifiable” predictions from the zero-shot classification technique revealing several nuances, including the lack of pertinent context in many of the narratives and the absence of appropriate categories for selection. For example, 57.3% of cases classified under the “part\_of\_body” column suggest that the narrative alone did not refer to specific body parts. Furthermore, in the “event\_type” column, multiple narratives suggest that an employee drowned; however, there is no variable for “drowning” in this category. For each database column, the left bar represents the original IMIS database null classification, and the right bar represents GPT-3.5 null classification. In this context the fields “other” and “nonclassifiable” are considered equivalent, since both indicate that a relevant entry does not exist in the database.

The model suggests new categories in some cases, but they are rarely chosen. For example, the LLM generated the following new categories for various fields.

- “event\_type”: struck-between, electrocution, drowning, etc.
- “env\_factor”: other, wind/weather condition, electric hazard, etc.
- “nature\_of\_inj”: crushing/crushed, laceration, struck-by vehicle, paralyzation, etc.
- “occ\_code”: truck drivers, iron workers, flaggers, bulldozer operators, etc.

|                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Road Construction Incident: #9</b>                                                                                                                                                                                                                         |
| Column: part_of_body                                                                                                                                                                                                                                          |
| Abstract: “On July 29, 2004, an employee working as a flagman during road paving was fatally struck by a tractor trailer.”                                                                                                                                    |
| Original Database: body system                                                                                                                                                                                                                                |
| LLM Classification: nonclassifiable                                                                                                                                                                                                                           |
| <b>Road Construction Incident: #70</b>                                                                                                                                                                                                                        |
| Column: occ_code                                                                                                                                                                                                                                              |
| Abstract: “On August 11, 2004, Employee #1 was pinned under a 25-ft long, 3,000 pound steel I-beam, which rolled off a pile of stacked beams. He was hospitalized with a fractured left leg.”                                                                 |
| Original Database: carpenters                                                                                                                                                                                                                                 |
| LLM Classification: construction laborers                                                                                                                                                                                                                     |
| <b>Road Construction Incident: #340</b>                                                                                                                                                                                                                       |
| Column: degree_of_inj                                                                                                                                                                                                                                         |
| Abstract: “Employee # 1 was operating a 1993 Caterpillar pneumatic tire asphalt compactor when he lost control of the compactor. The compactor rolled over twice, and the employee was pinned underneath the compactor. The employee was crushed and killed.” |
| Original Database: fatal                                                                                                                                                                                                                                      |
| LLM Classification: fatal                                                                                                                                                                                                                                     |

Fig. 9. Examples of the LLM classifying narrative (zero-shot classification).

- “part\_of\_body”: pelvic area/pelvis, head/face, torso, upper body, etc.

To enhance the model's effectiveness, we recommend iteratively updating the list of potential categories during inferencing, allowing the model to select the newly curated labels more frequently. If these new categories are selected over the existing ones, it may indicate the need to expand the category list. Moreover, it was found that replacing “other” with “nonclassifiable” in many fields improved evaluation outcomes, as the model tends to select “nonclassifiable” more frequently based on the prompt design.

### 3.2.2. Few-shot results

This classification aimed to highlight instances where the LLM can be leveraged to re-evaluate the initial database coding, possibly suggesting more suitable entries for specific incident categories. Unlike zero-shot classification, this approach tends to yield better performance due to the model possessing knowledge of the original database coding. Ultimately, this provides insights into the quality of the database by showcasing the model's ability to refine and propose more accurate classifications based on the existing information within the context of the individual abstracts. The two cases presented in Fig. 11 were selected to demonstrate our interpretation of the model's ability to impose a more well-informed categorical variable. The abstract for incident #20 deliberately claims that the employee collapsed due to asphyxia, whereas the original classification deemed otherwise. Similarly, the abstract for incident #186 also indicates this insight where the original classification does not correspond to the narrative in any allusive manner.

### 3.2.3. Overall performance

The performance metrics from Eqs. (8)–(11) were then calculated for both few-shot and zero-shot classification scenarios, as presented in Table 4. It was expected that GPT-3.5 would perform best when there are few categories to classify, as exemplified by the “degree\_of\_injury” column (Fig. 12). This category only has three potential categories based on what was originally coded in the database: fatal, hospitalized, and non-hospitalized, which would explain why it scored highest across all metrics. In many cases, the fields that achieved the highest accuracy (overall correctness) also achieved higher F1 scores (balanced true positive predictions). Similarly, if the predictions indicated poor performance, the accuracy and other provided metrics were typically uniformly lower. The bold and highlighted cells in this table show the highest and lowest-scoring predictions.

Aside from the degree of injury, where peak performance is consistently achieved, the performance of the other column predictions expands the insights that can be derived from this type of LLM classification. With the zero-shot predictions, the model obtained the highest accuracy when classifying the event type, which suggests that most narratives contained enough context to derive at least the type of event to the best of its ability. In contrast, the environmental factor, which appears to allude to any environmental hazards contributing to the accident, achieved the lowest across almost every metric for this prompting technique. This disconnection could be attributed to the isolated narrative not containing enough relevant information to appropriately categorize environmental contributions, or in other categories where performance was also low.

A deeper analysis of Table 4 reveals instances where the few-shot LLM prediction disagrees with the original classification, typically only in a small fraction of cases. There are four possible scenarios.

- Scenario 1. Coded classification more consistent with narrative than LLM classification
- Scenario 2. LLM classification more consistent with narrative than coded classification
- Scenario 3. Both classifications are consistent with narrative and no preference can be provided based on information provided

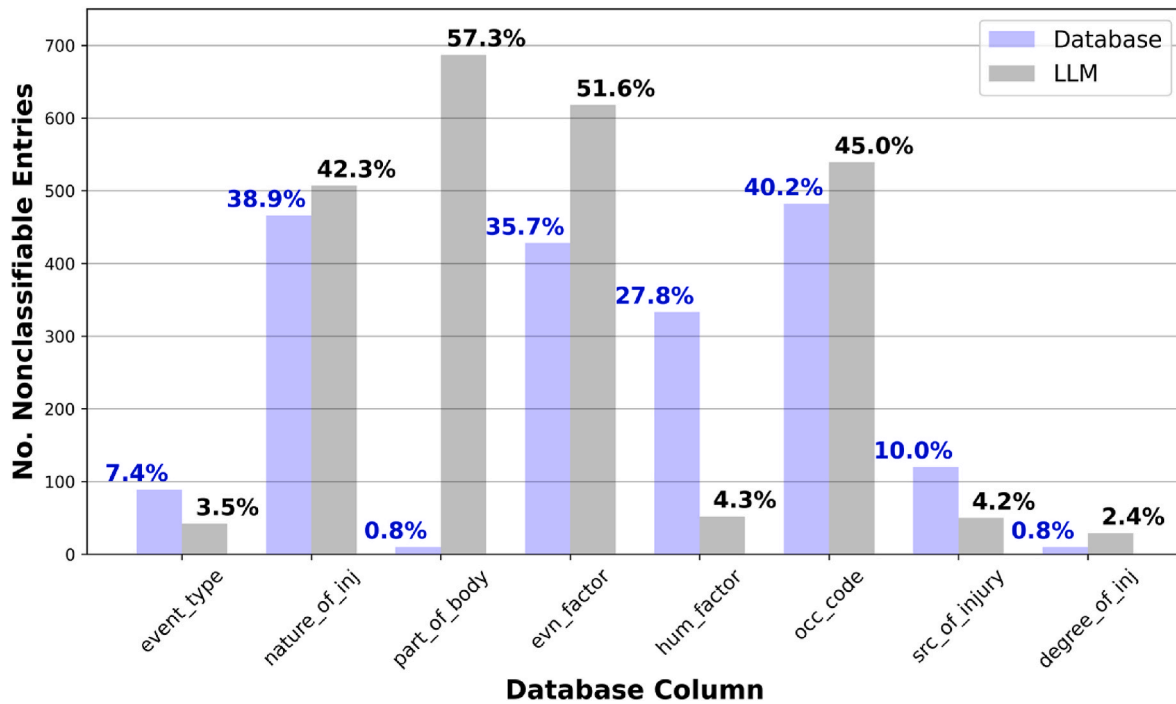


Fig. 10. Nonclassifiable entries with LLM zero-shot classification.

|                                                                                                                                                                                                                                                           |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Road Construction Incident: #20</b>                                                                                                                                                                                                                    |                 |
| Column: nature_of_inj                                                                                                                                                                                                                                     |                 |
| Abstract: "On February 19, 2004, Employee #1, self-employed contractor, was performing maintenance and repair work on a commercial building. Apparently he collapsed due to asphyxia or intoxication from paint vapors. Employee #1 was pronounced dead." |                 |
| Original Database:                                                                                                                                                                                                                                        | nonclassifiable |
| LLM Classification:                                                                                                                                                                                                                                       | asphyxia        |
| <b>Road Construction Incident: #186</b>                                                                                                                                                                                                                   |                 |
| Column: event_type                                                                                                                                                                                                                                        |                 |
| Abstract: "An employee was backing a hydraulic post driver into position to install metal posts for a new guardrail. He was electrocuted when the unit contacted a 7,200-volt overhead power line."                                                       |                 |
| Original Database:                                                                                                                                                                                                                                        | other           |
| LLM Classification:                                                                                                                                                                                                                                       | shock           |

Fig. 11. Examples of the LLM re-evaluating database coding (few-shot classification).

Scenario 4. Both LLM and original coding are inconsistent with the narrative

**Table 4**  
Performance evaluation of LLM classification.

| Column        | Few Shot     |              |               |              | Zero Shot    |              |               |              |
|---------------|--------------|--------------|---------------|--------------|--------------|--------------|---------------|--------------|
|               | Accuracy (%) | Recall (%)   | Precision (%) | F1Score (%)  | Accuracy (%) | Recall (%)   | Precision (%) | F1Score (%)  |
| event_type    | 93.57        | 97.48        | 95.89         | 96.68        | <b>64.36</b> | <b>75.22</b> | <b>81.67</b>  | <b>78.31</b> |
| nature_of_inj | 85.39        | <b>91.91</b> | 92.33         | 92.12        | 31.05        | 69.27        | 36.01         | 47.39        |
| part_of_body  | <b>84.64</b> | 94.41        | <b>89.10</b>  | <b>91.68</b> | 21.95        | 55.72        | 26.59         | 36.00        |
| evn_factor    | 92.65        | <b>98.06</b> | 94.39         | 96.19        | <b>15.69</b> | 43.82        | <b>19.64</b>  | <b>27.13</b> |
| hum_factor    | <b>94.16</b> | 97.66        | <b>96.33</b>  | <b>96.99</b> | 20.87        | <b>38.17</b> | 31.53         | 34.53        |
| occ_code      | 86.48        | 96.37        | 89.39         | 92.75        | 32.30        | 47.08        | 50.72         | 48.83        |
| src_of_injury | 89.98        | 94.07        | 95.40         | 94.73        | 43.75        | 64.84        | 57.05         | 60.70        |
| degree_of_inj | <b>98.91</b> | <b>99.66</b> | <b>99.25</b>  | <b>99.45</b> | <b>93.82</b> | <b>97.32</b> | <b>96.32</b>  | <b>96.81</b> |

Fig. 13 presents the results of a manual case-by-case investigation of "event\_type" where 77 cases (6.4%) were classified differently. The "event\_type" field was selected as the primary candidate for the manual evaluation because it represents the "what happened" of an accident – the most critical variable for identifying safety precursors and implementing preventative measures.

In Scenario 1, only 16.9% of cases found the original classification more consistently relevant to the narrative. In most inconsistencies (Scenario 2 – 37.7%), GPT-3.5 selected a more contextually relevant category. For example, as shown in Fig. 11, the narrative of accident #20 suggests that an employee collapsed due to asphyxia. In this case, the original classification was deemed "nonclassifiable"; however, the model selected "cardio-vascular/respiratory system failure." Additionally, 45.2% of the cases showed that both classifications were either contextually consistent (Scenario 3 – 20.8%) or neither was consistent (Scenario 4 – 24.6%) with the narrative text. This manual investigation can be conducted for the remaining categories to evaluate and validate few-shot classification. Currently, no automatic validation techniques have been established for this type of analysis. Future studies can explore new methods to assess model performance in classification tasks to alleviate the time-consuming and manual nature of this evaluation.

Furthermore, for the few-shot mode, the human factor column obtained the highest accuracy and F1 Score, indicating that the original coding was typically sufficient. The part of the body, however, achieved the lowest accuracy, which could be partly due to the narrative not



Fig. 12. LLM confusion matrix for “degree\_of\_injury” column in IMIS database.

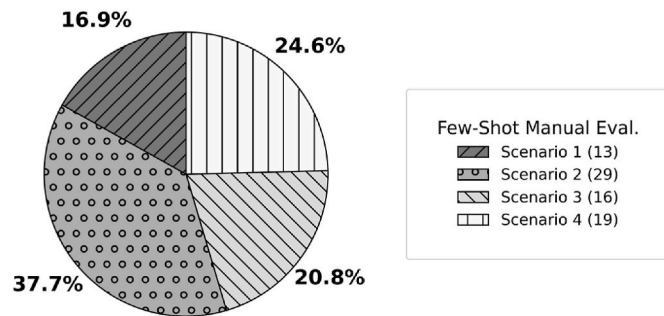


Fig. 13. Manual evaluation of “event\_type” few-shot classification results.

containing sufficient information to properly classify.

Inadvertently both of the prompting techniques reveal deeper insights into future analysis of this nature, with a strong emphasis on the quality of the incident abstracts. For the LLM to be used to its full potential, the narratives should encapsulate enough contextual information to be able to classify the incidents into various categorical variables. The lower performance obtained through the zero-shot prompting also emphasizes the imperative to have high-quality descriptions coded into the database to perform similar analyses.

### 3.3. Estimated model costs

The entire approach, including clustering, summarization, and classification, cost approximately \$2.33 USD (Table 5). This cost is relatively low considering the database contains over 1000 entries and almost one million tokens were processed by the models. The self-consistency evaluation require similar resources as the initial summary generation, which may only be required for validation and verification purposes. Significant inferencing was saved by batching multiple narratives at once (20 in this case) for classification and summarization purposes. Furthermore, since the few-shot mode outperforms zero-shot, the cost is negligible for these improvements. It is anticipated that these costs would scale linearly with larger datasets.

## 4. Conclusions

As fatalities and severe injuries persist in the construction industry, more comprehensive and advanced data-driven analyses are needed to understand their nature. The plethora of publicly available data on accidents mandated by OSHA and other federally run programs enables previously underutilized textual data to be explored through various NLP approaches. These range from traditional term frequency analyses to contextual analyses made possible by the transformer architecture in LLMs. This study proposes a modified approach to analyzing this textual data with GPT-3.5, a more performant model than others commonly used at the time of this study. Focusing specifically on the understudied highway sector of construction, which blankets a wide range of construction activities, the findings of this study aim to enhance the classification and identification of major categories of accidents within the industry.

Leveraging LLMs to perform cluster summarization allows for the automatic interpretation of underlying similarities between incidents, which previously required manual dissemination with general ML algorithms. The automated summarization of incidents within distinct clusters reveals dominant accident categories such as fall perils associated with cranes and aerial lifts or equipment misuse posing threats to employees. Leading accident types such as struck-by accidents and falls can be studied at a more fine-grained level, revealing subcategories that were not implicitly indicated by the original databases. Despite struck-by accidents comprising over 50% of incidents in the highway construction industry, the generated summaries reveal more nuanced subtypes within this category. Specific struck-by accidents, such as vehicle strikes beyond barriers, backing-up dump trucks, and heavy equipment run-overs, provide detailed insights beyond the broad “struck-by” categorization in the original database. Overall, an emphasis is placed on the specific techniques implemented, including MapReduce and SelfCheckGPT prompting techniques, which combined resulted in 83% self-consistent summaries with minimal hallucinations for the described groupings.

In addition to the outcomes of summarization, the role of LLMs as multi-class classifiers has immediate implications for supporting safety personnel in evaluating the quality of existing data. Demonstrating that these models can attain high classification performance through transfer and in-context learning, they offer insights not previously accessible

**Table 5**  
Estimated API cost breakdown by analysis component.

| Model & Price (USD)                                                           | Component                           | Process Breakdown                                                                                                                                                               | Est. In Tokens | Est. Out Tokens | Cost (USD)                            |
|-------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------|---------------------------------------|
| <b>ada-002</b><br>\$0.10/<br>M tok.                                           | Embeddings (one-time)               | 1198 narratives (~160k input tok. at ~136 tok. ea.)                                                                                                                             | 160,000        | ~               | <b>\$0.016</b>                        |
| <b>gpt-3.5-turbo</b><br>Input: \$0.50/<br>M tok.<br>Output: \$1.50/<br>M tok. | Map Summaries                       | 1198 narratives (~160k input tok. at ~136 tok. ea.)<br>Chunk summaries (~300 output tok./chunk)<br><i>Approx 124 chunks for 1198 narratives</i>                                 | 160,000        | 37,200          | <b>\$0.136</b>                        |
|                                                                               | Reduce Summaries (10 clusters)      | Chunk (fact) summaries (~300 input tok./chunk)<br>Consolidated summary (~200 output tok./cluster)<br><i>Intermediate &amp; final summaries after consistency check</i>          | 37,200         | 2000            | <b>\$0.043</b><br>(\$0.021/iteration) |
|                                                                               | Self-Consistency (10 clusters)      | Intermediate summary (~200 input tok./cluster)<br>Fact inputs (15 facts/chunk at ~500 input tok./chunk)<br>Fact outputs (15 facts/chunk at ~600 output tok./chunk)              | 64,000         | 74,400          | <b>\$0.144</b>                        |
|                                                                               | Classification (zero-shot, 8 tasks) | 1198 narratives (~160k input tok. at ~136 tok. ea.)<br>Classification prompts (~8k input tok.)<br>Output (~20 output tok./narrative)                                            | 168,000        | 23,960          | <b>\$0.959</b><br>(\$0.120/task)      |
|                                                                               | Classification (few-shot, 8 tasks)  | 1198 narratives (~160k input tok. at ~136 tok. ea.)<br>Classification prompts (~8k input tok.)<br>Shot examples (~18k input tok./example)<br>Output (~20 output tok./narrative) | 186,000        | 23,960          | <b>\$1.032</b><br>(\$0.129/task)      |

without manual effort. In this study, several examples showcase the ability to automatically reassess the information coded in the database across various text-based categories when given the existing entry (few-shot prompting). For instance, GPT-3.5 proved its ability to extract significant context from accident abstracts that explicitly include information about the type of injury sustained, i.e., electric shock, whereas the original classification deemed otherwise. In zero-shot prompting scenarios where the model classifies incidents into predefined categories without prior coded knowledge, the inherent quality of the abstracts can be assessed. The model tends to indicate “nonclassifiable” when there is insufficient information in the abstract, underscoring the importance of including comprehensive incident details in future reports. Furthermore, the LLM-based classification deviated from expected output in less than 2% of cases, aside from subtle duplication issues. A total of 314 entries resulted in hallucinated output labels, being more prominent in zero-shot mode; however, this is negligible considering the large number of tasks the model was prescribed.

It is imperative to acknowledge the limitations inherent in techniques such as the K-means clustering algorithm, which may not fully capture the intricacies of the data, especially in the presence of high-dimensional embeddings. Future studies should explore alternative clustering techniques and methods for the determination of an optimal number of accident types. In this study, the final number of clusters was determined qualitatively by reviewing a varying number of clusters, as there was no clear optimal structure under SSE and elbow method analyses. This discrepancy should be addressed in subsequent iterations through quantitative evaluations such as silhouette scores or Davis-Bouldin index. Additionally, the current study used manual expert evaluation to validate LLM responses which is expensive and time consuming. The development of automated assessment for generated content must be explored. The potential for refining and expanding upon these methods with the incorporation of cutting-edge language processing offers a promising avenue for future research in this domain.

As the field of NLP continues to evolve at an accelerated pace, driving advancements in embedding models and language proficiency, the trajectory suggests that future iterations will likely outperform current ones. The findings of this study, anchored in the capabilities of OpenAI's GPT-3.5, serve as a foundation for future exploration amidst the development of other pre-trained LLMs. Later initiatives that stem from this research should compare these results with other models and open-source alternatives to evaluate performance versus privacy trade-offs. Nonetheless, the current approach presents a low entry barrier for large-scale data processing within transportation agencies, whereas these models can perform labor-intensive tasks at substantially lower costs. While the specific outcomes of the analysis may vary with the adoption of new and improved models, the overarching theme remains one of providing high-level insight into targeted areas of concern in highway construction.

#### CRedit authorship contribution statement

**Mason Smetana:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lucio Salles de Salles:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology. **Lev Khazanovich:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

#### Declaration of generative AI and AI-assisted technologies in the writing process

OpenAI's GPT-3.5 (version: gpt-3.5-turbo-1106) was used to conduct the majority of analysis in terms of summarization and classification, as defined in the methodology. In addition, OpenAI's ChatGPT was used to assist proofreading and revision of the manuscript. The authors of this

paper acknowledge the limitations of LLMs such as potential biases, errors, and gaps in knowledge, as documented in the results of the study.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The accident data is publicly available on the United States Occupational Safety and Health Administration (OSHA) website: [https://enforcedata.dol.gov/views/data\\_summary.php](https://enforcedata.dol.gov/views/data_summary.php).

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